MICROBIOLOGY · MCQ SET

Poststreptococcal GN & IgA Nephropathy

SGL 2 · Prof. Dr. Ruqaya M. Al Barzinji · GUE BLOCK · PRACTICE MCQs

Microbiology GUE Block · MEDucation by Mattin Majid · Practice MCQs

20 questions, mixed difficulty, 8 case-based. Attempt every question before checking the answer key on the last page.

4	10	6
Easy	Medium	Hard

1. EASY

What type of mechanism defines glomerulonephritis (GN) as a disease category?

A. Direct toxic injury by bacterial endotoxin

B. An immunologic mechanism triggering inflammation and proliferation of glomerular tissue

C. Mechanical obstruction of the glomerular capillaries

D. Primary genetic structural collagen defect only

E. Ischemic injury from renal artery stenosis

2. EASY

Which organism precedes poststreptococcal glomerulonephritis (PSGN)?

A. Staphylococcus aureus

B. Nephritogenic strains of group A Streptococcus pyogenes

C. Escherichia coli

D. Enterococcus faecalis

E. Klebsiella pneumoniae

3. EASY

What is the other name for IgA nephropathy?

A. Goodpasture disease

B. Berger disease

C. Alport syndrome

D. Fabry disease

E. Wegener disease

4. EASY

Approximately how long after strep throat does PSGN classically develop?

A. 1-2 days

B. 10 days

C. 6 weeks

D. 6 months

E. 1 year

5. MEDIUM

What class of enzyme is SpeB (streptococcal pyogenic exotoxin B)?

A. A serine protease

B. A cysteine protease

C. A metalloprotease

D. A lipase

E. A nuclease

6. MEDIUM CASE

A child develops dark, tea-coloured urine, facial and hand swelling, and decreased urine output about 10 days after a documented episode of streptococcal pharyngitis.

What is the most likely diagnosis?

A. IgA nephropathy

B. Poststreptococcal glomerulonephritis

C. Minimal change disease

D. Diabetic nephropathy

E. Lupus nephritis

7. MEDIUM

Which molecular property allows SpeB to activate complement without requiring IgG?

A. Direct activation of the mannose-binding lectin (MBL) pathway

B. Direct binding to C1q

C. Cleavage of IgA1 hinge region glycans

D. Activation of the terminal complement complex only

E. Inhibition of complement regulatory proteins

8. MEDIUM

What is the key molecular defect in the IgA1 molecule that underlies IgA nephropathy?

A. Deficient galactosylation of O-linked glycans in the hinge region

B. A point mutation in the IgA1 constant region

C. Absent J-chain expression

D. Increased sialylation of the Fc region

E. Loss of the IgA1 hinge region entirely

9. MEDIUM CASE

A 20-year-old man has recurrent episodes of visible hematuria, each beginning 1-2 days after an upper respiratory tract infection, over several years.

What is the most likely diagnosis given this timing pattern?

A. PSGN, because of the clear infection trigger

B. IgA nephropathy, given the synpharyngitic pattern of recurrent hematuria

C. Minimal change disease

D. Acute interstitial nephritis

E. Goodpasture syndrome

10. MEDIUM

Which immunoglobulins/complement components are classically found on renal biopsy immunofluorescence in IgA nephropathy, and in what glomerular location?

A. IgA, IgG, IgM and C3 in the mesangium

B. IgG alone in a linear pattern along the GBM

C. IgE alone in the subepithelial space

D. C1q alone in the capillary loops

E. IgM alone in the mesangium

11. MEDIUM

What happens to circulating levels of IgA1-containing immune complexes as a consequence of altered O-glycosylation?

A. They are cleared faster by the liver

B. Hepatic clearance is reduced, raising circulating levels

C. They are entirely degraded in the bone marrow

D. They convert to IgG

E. They are filtered freely without any glomerular deposition

12. MEDIUM CASE

A 22-year-old woman with a strong family history of IgA nephropathy is found to have microscopic hematuria on a routine urinalysis.

Which of her risk factors described in the lecture is relevant to her diagnosis?

A. Her age and sex alone, since males are more affected than females

B. A family history of IgA nephropathy, since family history is a recognized risk factor

C. Her blood type

D. Her height and weight

E. Her occupation

13. HARD

Which receptor, expressed on myeloid cells, plays a crucial role in the clearance of IgA immune complexes, and what does its cross-linking trigger?

A. FcγRIII; antibody-dependent cellular cytotoxicity against tumor cells only

B. FcαRI/CD89; phagocytosis, superoxide generation, antibody-dependent cellular cytotoxicity and inflammatory mediator release

C. TLR4; direct antigen presentation to T cells

D. CR1; direct lysis of immune complexes

E. FcεRI; mast cell degranulation only

14. HARD CASE

A patient diagnosed with IgA nephropathy in his twenties asks about his long-term kidney prognosis.

Which complication should be specifically discussed given this trajectory?

A. Acute reversible kidney injury with full recovery expected in all cases

B. Progression to end-stage renal disease, which occurs in 30-40% of patients 20-30 years after first presentation

C. Immediate dialysis is required at diagnosis in all cases

D. Spontaneous resolution without any long-term risk

E. Conversion to PSGN over time

15. HARD

Which of the following mechanisms contributes to SpeB's capacity to damage the glomerular basement membrane (GBM), independent of classical antibody-mediated injury?

A. Direct lysis of podocytes by pore formation

B. Plasmin-binding properties that facilitate complement activation and may cause proteolysis of the GBM

C. Direct binding to type IV collagen as an autoantigen

D. Inhibition of mesangial cell proliferation

E. Suppression of the alternative complement pathway

16. HARD CASE

A child recovering from PSGN has a renal biopsy showing some subendothelial immune deposits.

What underlying process best explains her 'subendothelial deposits' on biopsy despite SpeB being an exogenous bacterial antigen?

A. Direct deposition of streptococcal DNA

B. SpeB's molecular mimicry with endothelial cells

C. An unrelated autoimmune process

D. Random nonspecific protein trapping

E. IgA1 hinge-region glycosylation defects

17. HARD

Which combination of findings in IgA nephropathy's pathogenesis constitutes the described 4-hit model, in correct sequential order?

A. Increased Gd-IgA1 → anti-glycan antibody production → immune complex formation → nephron injury

B. Nephron injury → immune complex formation → anti-glycan antibody production → increased Gd-IgA1

C. Immune complex formation → increased Gd-IgA1 → nephron injury → anti-glycan antibody production

D. Anti-glycan antibody production → nephron injury → increased Gd-IgA1 → immune complex formation

E. All four hits occur simultaneously with no defined sequence

18. HARD CASE

A patient has recurrent macroscopic hematuria, with serum polymeric IgA1 mildly elevated, and the clinical team needs definitive diagnostic confirmation.

What is the most important diagnostic test to confirm IgA nephropathy in this patient?

A. Serum complement C3 and C4 levels alone

B. Renal biopsy with immunofluorescence/ immunohistochemistry

C. 24-hour urine protein alone

D. Throat culture for group A Streptococcus

E. Antistreptolysin O (ASO) titer alone

19. HARD

Which of the following non-renal conditions has been associated with IgA nephropathy, according to the lecture?

A. Coeliac disease, hepatitis, cirrhosis and HIV

B. Hypothyroidism only

C. Type 1 diabetes only

D. Primary hyperparathyroidism only

E. Sickle cell disease only

20. HARD CASE

A child with PSGN is found to have complement activation on laboratory testing that appears to occur even in the absence of significant IgG deposition.

Which single antigen is most specifically implicated in driving this finding, and by what pathway does it act?

A. GD-IgA1, acting via the classical complement pathway

B. SpeB, acting via the mannose-binding lectin pathway independent of IgG

C. Streptolysin O, acting via direct membrane lysis

D. M protein, acting via molecular mimicry with cardiac myosin

E. DNase B, acting via nucleic acid degradation

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	B	GN is defined as a set of renal diseases in which an immunologic mechanism triggers inflammation and proliferation of glomerular tissue.
2	B	PSGN follows infection with nephritogenic strains of group A Streptococcus pyogenes.
3	B	IgA nephropathy is also called Berger disease, after the physician who first described it.
4	B	PSGN usually occurs about 10 days after strep throat or scarlet fever.
5	B	SpeB is a cysteine protease that cleaves various host factors and acts as a nephritogenic antigen.
6	B	Dark urine, edema and oliguria about 10 days after streptococcal pharyngitis is the classic PSGN presentation.
7	A	SpeB can activate complement directly through the mannose-binding lectin (MBL) pathway, independent of IgG.
8	A	The key defect is deficient galactosylation of the O-linked glycans in the IgA1 hinge region, producing galactose-deficient IgA1 (GD-IgA1).
9	B	Recurrent hematuria occurring 1-2 days after infection (synpharyngitic) is characteristic of IgA nephropathy, distinguishing it from the longer latency of PSGN.
10	A	IgA nephropathy is characterised by mesangial deposits of IgA, IgG, IgM and C3.
11	B	Antibodies bound to O-linked glycans in the hinge region reduce hepatic clearance of circulating immune complexes, raising their levels.
12	B	Family history is explicitly stated as a risk factor for IgA nephropathy.
13	B	FcαRI/CD89 on myeloid cells is crucial for clearance of IgA complexes; its cross-linking triggers phagocytosis, superoxide generation, antibody-dependent cellular cytotoxicity and inflammatory mediator release.
14	B	30-40% of IgA nephropathy patients reach end-stage renal disease 20-30 years after their first clinical presentation — the key long-term prognosis point.
15	B	SpeB's plasmin-binding properties facilitate complement activation and may cause proteolysis of the glomerular basement membrane, independent of classical antibody-mediated mechanisms.
16	B	SpeB exhibits molecular mimicry with endothelial cells, which explains the presence of some subendothelial deposits in PSGN despite SpeB being an exogenous bacterial antigen.
17	A	The 4-hit model proceeds: increased circulating Gd-IgA1 → production of anti-glycan antibodies → immune complex formation (in circulation or in situ) → immune complexes in the mesangium causing nephron injury.
18	B	Renal biopsy with immunofluorescence/immunohistochemistry is the cornerstone of IgA nephropathy diagnosis.
19	A	The lecture specifically lists coeliac disease, hepatitis, cirrhosis and HIV as conditions associated with IgA nephropathy.
20	B	SpeB can activate complement directly via the mannose-binding lectin pathway, independent of IgG — explaining complement activation without significant IgG deposition in PSGN.